



Magic and ritual as affiliative strategies in response to ostracism threat[☆]

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ABSTRACT

The psychological motivations underlying social learning are contested. The Bifocal Stance Theory (BST; Jagiello et al., 2022a; Whitehouse, 2021) posits that individuals adopt either a ritual stance, which prioritizes social affiliation and norm adherence, or an instrumental stance, which emphasizes efficiency in achieving goal-directed outcomes. We hypothesize that actions can be viewed by social learners through the lens of either stance depending on a variety of contextual cues. To test this hypothesis, we employed an online adaptation of the pen-and-cups paradigm (Leighton et al., 2010), in which participants (recruited from *Prolific*, $N = 176$) repeatedly observed and reproduced modeled actions. They were randomly assigned to one of four conditions (*clear-frame*, *expert-frame*, *magic-frame*, or *ritual-frame*), systematically manipulating verbal frames, causal opacity, goal demotion, and the resolvability of opacity (i.e. whether the causal structure was in principle knowable). The Cyberball task (a collective ball-passing activity priming ostracism threat; Williams & Jarvis, 2006) was used to investigate whether affiliative motivations influence imitative fidelity. We find that individuals experiencing ostracism exhibit higher copying fidelity for actions that are causally opaque in irrevocable ways. We argue that when actions are interpreted and transmitted as “rituals” and “magic”, they serve social and affiliative functions, reinforcing group cohesion and cooperation rather than merely supporting the transmission of technically useful information. These findings challenge and extend existing accounts of cultural learning which argue that cultural transmission is primarily a matter of instrumentally directed learning. Because the design used experimentally supplied action frames, the findings should be read as a controlled test of threatened belonging on cue-guided learning, rather than as a naturalistic simulation of group formation. We argue that both affiliative and instrumental perspectives play a central role in the psychology of social learning.

1. Introduction

Human cultural evolution requires both mutation and persistence. Across domains as diverse as technology and ceremonial life, the transmission of cultural practices entails both innovations and high-fidelity copying. Extensive work has addressed the processes supporting cumulative culture and technological improvement (Caldwell, Renner, & Atkinson, 2018; Cavalli-Sforza & Feldman, 1981; Mesoudi & Thornton, 2018; Tennie, Call, & Tomasello, 2009) as well as the psychological mechanisms that govern the acquisition of instrumental skills (Lyons, Damrosch, Lin, Macris, & Keil, 2011; Nielsen, 2006; Nielsen & Blank, 2011). This body of work has yielded powerful accounts of how learners identify effective means–end relationships, selectively retain successful variants, and drive innovation over generations.

Nevertheless, not all cultural practices have a knowable causal

structure. For example, although magical procedures are oriented toward a desired outcome, they do not rely on procedures whose causal structure is fully specifiable. In some cases this causal opacity invites innovation and turnover (Hong, Slingerland and Henrich, 2022), while in other instances practices persist over long periods of time and are transmitted with striking fidelity (Whitehouse, 2021). It may be tempting to view magical practices through the same lens as we would view complex forms of technology, which are likewise often causally opaque to users. However, we argue that attention to the persistence of rituals, conventions and other practices with irresolvably opaque means–end relationships usefully complements approaches that treat social learning as oriented toward efficiency and outcome optimization. More broadly, understanding why such practices remain stable, and why they are sometimes transmitted with greater fidelity than demonstrably effective technologies, requires a framework that goes beyond instrumental reasoning alone. Specifically, an adequate account of cultural

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transmission must specify the conditions under which learners interpret observed actions, evaluate their causal structure, and attend to surrounding social and contextual cues in ways that differentially promote innovation or faithful reproduction.

1.1. Magic as engineering and affiliating

A prominent account within anthropology and cultural evolution has been to propose that magical practices are instrumentally motivated responses to fitness-relevant problems, such as illness, misfortune or environmental volatility (Dubourg, Fitouchi, & Baumard, 2022; Hong, 2022; Hong & Henrich, 2021; Singh, 2021, 2022). According to such accounts, ritual and magical practices only appear non-instrumental to outside observers; practitioners themselves are assumed to pursue them as an unambiguously goal-directed effort to achieve desired physical outcomes. In addition to classic anthropological accounts of magical practices among the Azande and Trobriand Islanders (Evans-Pritchard, 1937; Malinowski, 1945), more recent ethnographic analyses of rain-making, divination, shamanism, and witchcraft document cases in which practitioners reason explicitly about causal efficacy, discard variants perceived as ineffective and innovate in search of improved outcomes (Boyer, 2020; Hong, Slingerland and Henrich, 2022; Singh, 2017). These findings seem to support the conclusion that rituals and magical practices are transmitted through the same instrumental reasoning processes that govern complex technologies and cumulative culture. On this view, cultural stability is a product of error management, while innovation results from efforts to improve their causal efficacy. This position would leave little room for consideration of affiliative and normative motivations in cultural transmission.

By contrast, a substantial body of research suggests that ritual practices are adopted not because they are effective tools in controlling material outcomes, but because they play a central role in regulating social life and sustaining prosocial commitment and cooperation. Across cultures, rituals serve diverse social functions, including signalling commitment and mate quality (Singh & Bronstad, 1997) or commemorating shared values and histories (Kertzer, 1989). Empirical and theoretical work has strengthened this view by showing that ritual participation can affect cooperation, bonding and commitment through several partly overlapping pathways (Fischer, Callander, Reddish, & Bulbulia, 2013; Lang & Kundt, 2024; Rossano, 2020). One prominent pathway involves social synchrony, whereby coordinated activities such as rhythmic movement, chanting, singing or marching reliably increase cooperation among participants (Catmur & Heyes, 2013; Fessler & Holbrook, 2016; Hagen & Bryant, 2003; Hove & Risen, 2009; Mehr, Krasnow, Bryant, & Hagen, 2020; Mogan, Fischer, & Bulbulia, 2017; Reddish, Fischer, & Bulbulia, 2013; Wilermuth & Heath, 2009). For instance, field evidence suggests that rituals involving synchronous movement enhance pro-social attitudes and, under some conditions, cooperative behaviour in economic games (Fischer et al., 2013), while also aligning physiological arousal among performers and emotionally involved spectators, thus not limiting ritual effects to those carrying out the actions themselves (Konvalinka et al., 2011).

Rituals also frequently require substantial investments of time, effort and material resources while producing no obvious technical output. These costs allow participation to function as a signal of commitment to the group, as engagement in such practices is difficult to reconcile with indifference or hostility toward the collective (Irons, 2001; Sosis & Alcorta, 2003). Extreme forms of participation in rituals are associated with increased prosocial behaviour toward the religious community (Xygalatas et al., 2013) and the signalling dynamics of such costly participation are influenced by participants' positions within their local social hierarchies (Xygalatas et al., 2021). In this way, costs reinforce group attachment, in part through dissonance-reducing processes that motivate positive attitudes toward the group following effortful initiation (Aronson & Mills, 1959).

A further line of work treats ritual as a mechanism for coordinating collective action and stabilizing cooperation. On this view, rituals help mobilize psychological and social resources such as commitment, trust, shared attention, and group-level coordination, while also functioning as cooperative signals in mutualistic social systems (Lang & Kundt, 2024). Related accounts emphasize how rituals act as identity markers undergirding group identification and in some cases (e.g. life-changing initiation rituals) may give rise to personally transformative shared experiences leading to identity fusion – a visceral feeling of oneness with the group, motivating extreme forms of pro-group action (Whitehouse, 2018). Equally, ritual frequency and emotional intensity have been shown to shape the scale and strength of social bonding (Whitehouse, 2004), further illustrating how cultural practices generate enduring social structures outside of the realm of physical causation.

Taken together, these approaches converge on the notion that rituals do not persist because they are efficient interventions in the material world, but because they organize attention, commitment, identity, and cooperation in social groups. There is little reason to copy irresolvably opaque (and thus instrumentally inefficient) behaviours unless one wishes to affiliate with a group that has adopted them. This would help to explain why these practices survive the winnowing effects of cultural evolution. That is, they are not easily replaced by actions construed in instrumental terms, as they fulfil a distinct function regulating patterns of cooperation.

In light of these observations, some of the assumptions underlying mostly goal-driven accounts of magic in particular (and ritual practices more broadly) may require some rethinking (Whitehouse, 2023). First, while some practices show demonstrably high rates of innovation (Hong, Slingerland and Henrich, 2022), others persist with remarkable stability even when their causal inefficacy is readily observable across time (Whitehouse, 2011). Furthermore, accounts that typically assume that the presence of an end-goal deterministically elicits instrumental reasoning do not address why goal-directed yet causally opaque practices seem to be sometimes transmitted like technology (Hong, Slingerland and Henrich, 2022) and sometimes like ritual (Whitehouse, 2021). Both cases seem to readily co-exist, which raises a fundamental question: what determines whether a learner treats a magical practice as a problem-solving activity or as a normative and socially meaningful tradition?

1.2. The bifocal stance theory

The Bifocal Stance Theory (hereafter BST; Fig. 1) was developed in part to address this puzzle (Jagiello, Heyes, & Whitehouse, 2022a; Whitehouse, 2021). Rather than assuming a single and predominantly instrumental mode of cultural learning, BST proposes that humans flexibly adopt different motivational stances when observing and copying the actions of suitable models. Specifically, learners may approach the same behavioural sequence through either an *instrumental stance* or a *ritual stance*, depending on how the action is interpreted. This distinction dispenses with clear-cut action taxonomies which are unlikely to capture “rituals”, “magic” and “technology” in ways that are objectively applicable—after all, what seems “magical” to one person might appear quite “technological” to someone else, and vice-versa (Jagiello, Heyes, & Whitehouse, 2022b). Accordingly, the study design on which we report here uses experimentally framed action sequences that vary in the degree to which they cue conventionality or instrumentality – testing how these cue combinations bias learners' interpretations of observed action. When adopting an *instrumental stance*, learners interpret a given action as a means to an end, assuming that its causal structure is, at least in principle, knowable. When adopting this stance, outcomes are prioritized, resulting in lower-fidelity copying and greater openness to innovation. By contrast, when adopting a *ritual stance*, there is no better way to perform the action than that which is stipulated by convention and buttressed by norms and institutions. Thus,

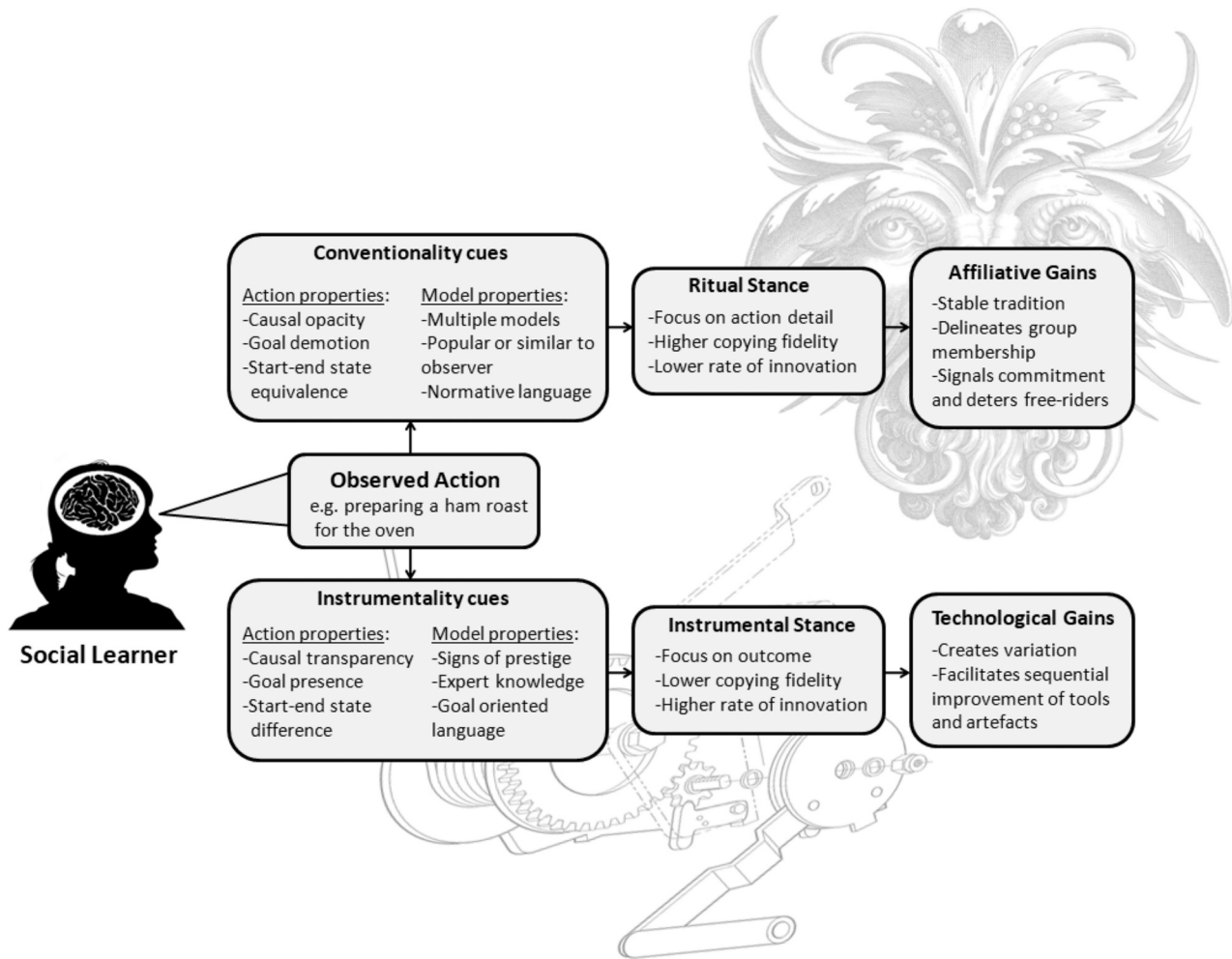


Fig. 1. Schematic overview of the Bifocal Stance Theory (from Jagiello et al., 2022a).

the ritual stance promotes high-fidelity copying and resistance to deviation as any component of the action may be socially required. Importantly, the ritual stance is not identical to conventional action: the latter refers to features of the action or its framing, whereas the former refers to the learner's interpretive orientation toward it.

Rather than adhering to a taxonomy of action (technology vs. ritual), we therefore propose a taxonomy of motivation (instrumental vs. affiliative) which describes social learning as a process that is both *observer-dependent* and *cue-sensitive*. By that we mean that a learner's interpretation of an observed action is shaped not only by the features of the action itself (such as *causal opacity* or *goal-demotion*; Kapitány & Nielsen, 2019; Nielsen, Tomaselli, & Kapitány, 2018) but also by a series of contextual cues which, for instance, may signal group affiliation through the use of certain verbal frames (Clegg & Legare, 2016a, Clegg & Legare, 2016b) or characteristics of those modelling the behaviour (Herrmann, Legare, Harris, & Whitehouse, 2013). In particular, we argue that the number of cues related to group membership, normativity and social inclusion (relative to the number of cues that signal instrumentality) may direct learners toward a ritual stance, even in cases when specific end-goals are made salient (as is often the case with “magic”), thus increasing copying fidelity for purposes of affiliation. Overall, our empirical efforts concern learners' interpretive stances rather than ontologically fixed action kinds, as we manipulate exposure to cue bundles in order to test how they guide interpretation and transmission.

The notion that social learning might be governed by instrumental and affiliative motivations has been discussed in the developmental literature for some time (Over & Carpenter, 2013; Uzgiris, 1981).

Indeed, our predictions are primarily based on initial research showing that third-party ostracism as well as more direct experiences of exclusion prompt children to copy causally opaque actions with especially high fidelity following social threats (Watson-Jones, Legare, Whitehouse, & Clegg, 2014; Watson-Jones, Whitehouse, & Legare, 2016). To prime ostracism directly, Watson-Jones et al. (2016) used the Cyberball paradigm, which is a virtual ball-tossing task in which participants are either excluded (ostracism) or not (control). Their findings show that children who were excluded by members of their own ingroup displayed the highest fidelity in copying a causally opaque action, exceeding the fidelity of those not exposed to ostracism. This suggests that even mild experiences of social exclusion can meaningfully shape how modeled behaviour is processed and attended to, with potentially far-reaching implications for social learning. For the present study, this prior work is important because it motivates our central prediction: if ostracism heightens attention to socially meaningful or conventionally framed behaviour, then ostracism threat should selectively increase copying fidelity for actions that are more likely to be interpreted through a ritual stance, rather than for actions approached in purely instrumental terms. Accordingly, we use the term ‘ostracism threat’ to refer to a threat to social belonging, and ‘desire to affiliate’ to refer to the motivational response to that threat, namely an increased motivation to secure belonging through socially appropriate behaviour.

Moreover, these findings fit into the wider literature which documents children's remarkable sensitivity toward norm adherence (Kenward, Karlsson, & Persson, 2011; Keupp, Behne, & Rakoczy, 2013), arbitrary rules during pretend-play (Rakoczy, Warneken, & Tomasello,

2009; Schmidt, Rakoczy, & Tomasello, 2011), and their ability to knowingly commit to actions that are ineffective in achieving a desired end-state (overimitation; Buttelmann, Zmyj, Daum, & Carpenter, 2013; Corriveau et al., 2017; Horner & Whiten, 2005). In light of this research, it appears to us that the role of affiliative motivations should be considered alongside instrumental reasoning in order to make sense of social learning and, by extension, cultural transmission. Further, the preponderance of developmental data also stresses the importance to go beyond research that is exclusively conducted on child populations and relatively simple action paradigms. This would allow us to address questions about the generality of these effects in adults.

1.3. Resolvability and the cultural perception framework

BST is a learner-centric account in which stance adoption is shaped by multiple types of cues (Fig. 1), including verbal frames, model characteristics, action characteristics, social context and group standing. The Cultural Perception Framework (CPF) focuses on one subset of these cues: those concerning the perceived structure of the observed action itself (Fig. 2). Specifically, the CPF describes how learners may interpret action characteristics based on three core dimensions: the presence or absence of a salient end-goal, the degree of causal opacity, and the presumed *resolvability* of that opacity. It is therefore not intended as a complete model of stance adoption, but as a way of organizing the action-side cues that may push learners toward different interpretive frames. While all three dimensions are important to the framework, previous work has already examined the roles of goal-demotion and causal opacity more directly (e.g. Kapitány & Nielsen, 2017). Thus, the emphasis of the present study is on cues intended to approximate perceived resolvability.

An action is *causally opaque* when the relationship between its components and its outcome is not perceived by the learner to be transparently governed by physical causation (Kapitány & Nielsen, 2019; Legare, Wen, Herrmann, & Whitehouse, 2015; Whitehouse, 2011). However, we argue that opacity alone does not determine whether an action is viewed through an instrumental or a ritual lens. Many procedures that we view as a complex technology (such as computer chip assembly or chemical engineering) are opaque to naïve

observers, but are widely assumed to be causally resolvable given sufficient expertise and are thus more likely to cue instrumental reasoning. *Resolvability* refers to whether a learner perceives the causal structure of an action to be knowable within a physical causal framework, even if they themselves lack the requisite knowledge (Legare & Herrmann, 2013; Legare & Souza, 2012; Legare & Souza, 2014; Whitehouse, 2011). By contrast, actions that are causally opaque in ways that are perceived as *irresolvable*—that is, not plausibly grounded in physical causation—are predicted to favour a ritual stance as they may be interpreted as a part of the repertoire of practices that promote group cohesion. Importantly, we do not assume that perceived resolvability is universal and cross-culturally invariant. Whether, and under what conditions, learners treat opacity as resolvable is likely shaped by cultural experience, locally available expertise and background assumptions about what kinds of causal processes are possible.

We also note that, because the CPF focuses on action-feature cues, it should not be read as a mutually exclusive decision tree of activities as a whole. Many practices contain both instrumental and conventional dimensions. Sports provide a clear example: technique, training, and equipment may invite instrumental reasoning, whereas rules, uniforms, ceremonies, and etiquette may invite conventional or ritualized interpretations. The relevant question is therefore not whether an entire activity is interpreted as “instrumental” or “ritual,” but which features of that activity are made salient to the learner and how those features are weighted alongside other BST-relevant cues.

To date, research on the cognitive foundations of ritual (Kapitány & Nielsen, 2015, 2017) as well as previous efforts based on stance-related theorizing (Watson-Jones et al., 2014; Watson-Jones et al., 2016) have not manipulated *resolvability* in systematic ways, despite having discussed its importance in the past (Legare & Nielsen, 2015). This relative lack of direct experimental attention is one reason why resolvability is a focal dimension in the present study. Our approach seeks to capture the disambiguation of practices that are goal-focused but also causally opaque in either *resolvable* or *irresolvable* ways. Moreover, previously cited ethnographic research has not considered the concept of *resolvability* (Hong, Slingerland and Henrich, 2022; Singh, 2022), perhaps implicitly assuming that all practices are viewed as resolvable by their practitioners. We view this assumption to be an outstanding empirical

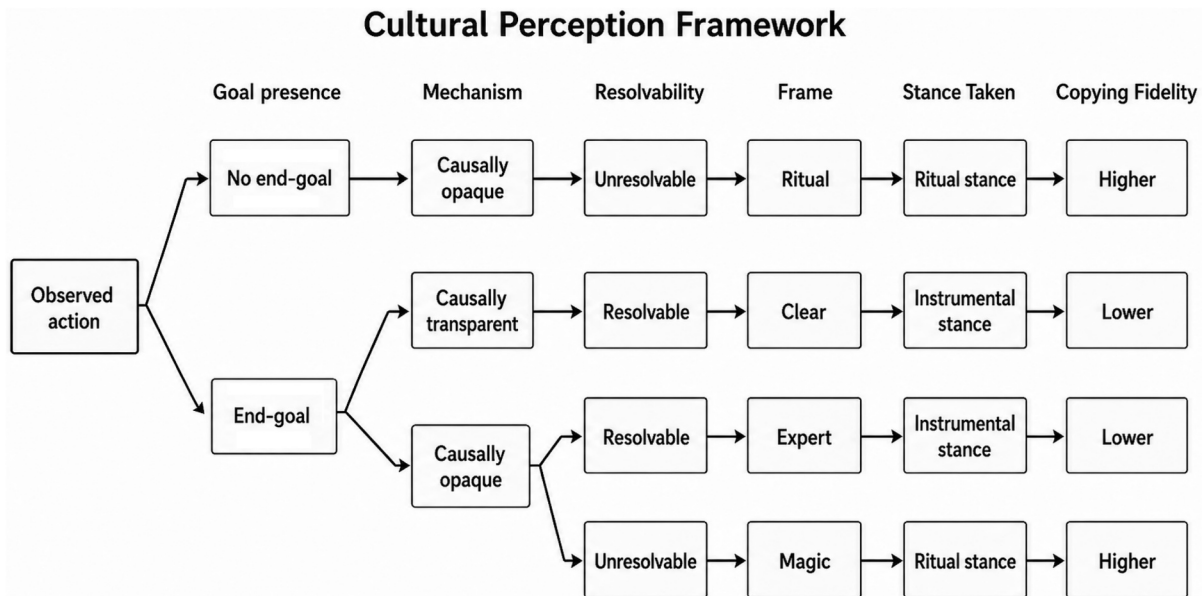


Fig. 2. The Cultural Perception Framework (adapted from Jagiello et al., 2022a) depicting the action-feature component of BST. Different paths outline how observed actions may be interpreted in terms of goal presence, mechanism and perceived resolvability, resulting in different framing profiles (ritual, clear, expert, magic) which map onto the conditions used in our imitation paradigm. The figure describes only the subset of BST-relevant cues concerning action interpretation; other cues relevant to stance adoption, such as verbal framing, model characteristics, and group standing, are represented in the broader BST account (section 1.2.).

question that is a central issue to address in order to understand the psychological motivations underlying cultural transmission. To fill these gaps, new experiments are needed to explore the pathways outlined in Fig. 2 more comprehensively and rigorously.

1.4. The level of deliberateness

In addition to attempting to reconcile different accounts of ritual research into the same framework, our learner-centric approach may help disentangle explicit goal attributions from the implicit psychological mechanisms that stabilize cultural practices (Jagiello et al., 2022a). A practitioner may consciously believe that a particular sequence of actions produces rain or heals the sick, yet still acquire and reproduce those actions via learned associations between norm adherence and affiliative reward rather than outcome monitoring.

Much of human cognition relies on slow, effortful and deliberative (type 2) or fast, automatic and associative processes (type 1; Evans & Stanovich, 2013; Kahneman, 2003). On a deliberative account, learners consciously engage in mentalizing, causal reasoning, and evaluation to assess what an action is for, whether its causal structure is understood and how it ought to be reproduced (Csibra & Gergely, 2009). On an automatic account, stance adoption would not be chosen but triggered by observable features of the action and its context. This process would operate outside of introspective access, not unlike other psychological effects, such as spatial compatibility, in which responses are faster when stimulus and response locations align (Lu & Proctor, 1995) or Stroop effects, in which word meaning interferes with colour naming despite explicit task goals (MacLeod, 1991). Distinguishing when stance selection reflects deliberative reasoning and when it reflects automatic triggering is therefore a key priority for future research, and motivates another feature of our present design which records learners' explicit justifications during imitation. While overlap between reported motivation and patterns of copying fidelity would suggest a conscious stance adoption, a poor match would favour the notion that stances are triggered automatically and that statements are post-hoc rationalizations that do not need to align with underlying associations.

2. Imitation study

The experiment reported here was designed to test core predictions of BST in adults, with a particular focus on how learners respond to goal-directed yet causally opaque action sequences when those sequences are framed with different combinations of instrumentality and conventionality cues (see Figs. 1 and 2). The study was preregistered (<https://osf.io/8bz9k>).

A central issue motivating the design is that goal-directed yet causally opaque practices do not always appear to be learned in the same way as causally transparent ones. As discussed above, some accounts emphasize that magical practices are best understood as instrumentally motivated attempts to solve practical problems (Dubourg et al., 2022; Hong, 2022). On such views, the presence of an explicit end-goal should strongly bias learners toward instrumental reasoning, even when the causal mechanism is opaque. This implies that goal-directed actions should primarily be approached as means to an end and copied in light of their presumed efficacy.

BST makes a different prediction. We propose that the presence of an end-goal is not, on its own, sufficient to determine how an action is learned. Instead, learners may weigh multiple cues, including goal salience, causal opacity, framing and whether the opacity is treated as resolvable or irresolvable. As a result, even actions with explicit end-goals may be copied with high fidelity when surrounding cues bias learners toward a ritual stance rather than an instrumental one. This is especially important under ostracism which we experimentally use here to trigger a desire to affiliate. If ostracism increases sensitivity to normatively relevant and affiliation-bearing behaviour, then it should selectively amplify copying fidelity for action sequences interpreted in

more conventional terms, rather than for those approached in purely instrumental terms.

To test our framework, we adapted the pen-and-cups task (Bird, Brindley, Leighton, & Heyes, 2007; Leighton, Bird, & Heyes, 2010) for use in an online adult sample and contrasted four framing conditions: a *clear-frame*, *expert-frame*, *magic-frame*, and *ritual-frame* condition. These conditions varied the salience of an end-goal, the degree of causal opacity, and cues intended to approximate whether the action's opacity was treated as resolvable or irresolvable. This made it possible to compare actions that were goal-directed but opaque in ways framed as resolvable with actions that were goal-directed but opaque in ways framed as irresolvable, while holding the motor structure of the task constant.

Hypotheses:

H1. Overall, copying fidelity is highest when no physical goal is present and whenever the shown sequence is causally opaque in irresolvable ways (*ritual-frame* condition). When sequences have a salient end-goal reached via irresolvable causal opacity (*magic-frame* condition), individuals copy those with higher fidelity than sequences which are either reached in causally opaque ways that are deemed resolvable (*expert-frame* condition) or that are completely transparent (*clear-frame* condition). The predicted order of copying fidelity therefore is *ritual* > *magic* > *expert* > *clear*.

H2a. When compared to controls, ostracized participants copy ritualistic actions (*ritual-frame* condition) more accurately.

H2b. Ostracized individuals copy magical actions (*magic-frame* condition) more accurately than controls. The difference is less pronounced than in the ritual condition.

H2c. Ostracism increases the copying fidelity for actions that are resolvably opaque (*expert-frame* condition) but less so than in the ritual and magic conditions.

H2d. Ostracism has no effect on the fidelity with which causally transparent actions (*clear-frame* condition) are copied.

H3. The degree to which participants reported feelings of rejection is predictive of their copying fidelity for ritualistic, magical and resolvably opaque actions but not for causally transparent actions.

H4. The ability to mentally picture and immerse oneself in the hypothetical scenario of re-enacting historical practices correlates positively with the effects of ostracism and causal opacity in a way where ostracized subjects with high imagery scores show the highest levels of fidelity when copying causally opaque practices.

Taken together, H2a–H2d express the key contrast between BST and more exclusively goal-focused accounts which make no predictions about ostracism and affiliative intent. If an explicit end-goal were sufficient in eliciting instrumental learning, then the goal-directed conditions should not differ from one another in terms of copying fidelity. BST instead predicts that the same end-goal can be approached differently depending on whether learners interpret the action through a more instrumental or more ritual lens, and that ostracism should selectively strengthen high-fidelity copying where conventionality cues outweigh cues of instrumentality.

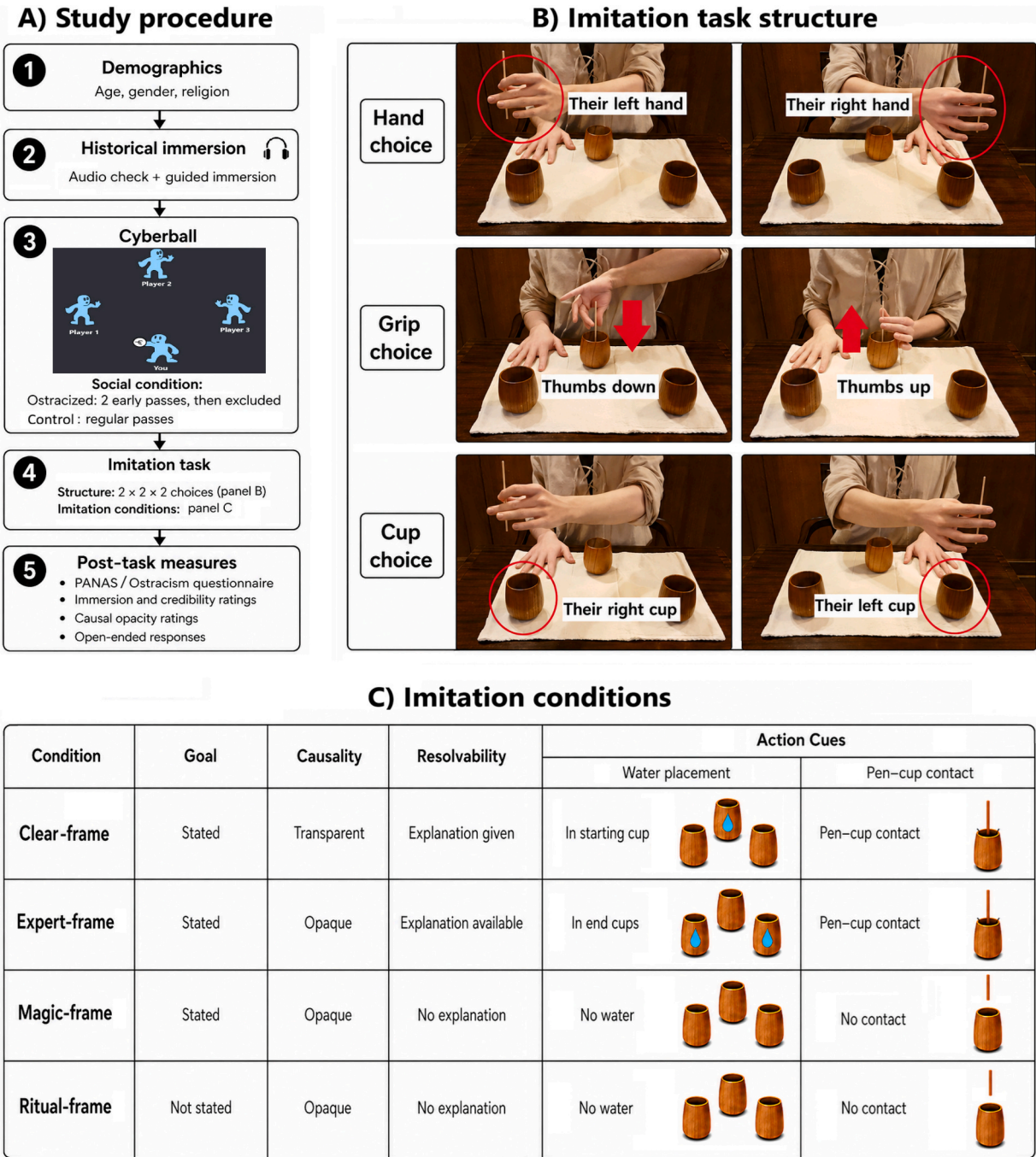
2.1. Participants

Sample size was determined a priori via preregistered G*Power calculations assuming a medium effect size in a 2×4 between-subjects design, resulting in a target sample of $N = 176$ (full input parameters are reported in the Supplementary Materials, F1). Participants from the UK's general population were recruited on Prolific.co and offered £6 for about 40 min of their time. The study was primarily advertised to users that indicated interest in 'Literature'. This was a pragmatic recruitment choice rather than a theoretically motivated decision: Participation in

our study involved an immersive cover story, so we used this preference as a proxy for recruiting participants who are more receptive to our materials. All participants were over the age of 18 and had normal or corrected-to-normal vision. Recruitment resulted in a sample of $N = 176$ (109 male; $M_{age} = 37.92$; $SD_{age} = 11.96$; 22 per between-subject condition). The study elements were embedded on Qualtrics and Pavlovia. Upon inspection of the submitted video files, three subjects did not perform the task. An additional three participants were recruited to meet the targeted sample size.

2.2. Ethics and open science statements

Participants were provided with a detailed consent form outlining the study procedure. The study was formally approved by the School of Anthropology and Museum Ethnography Research Ethics Committee. Great care was taken to debrief participants that exclusion during participation was manipulated and determined by chance alone. Additionally, subjects were informed during debrief that the sources and action sequences used in the study were fictional for the purpose of testing, rather than constituting actual practices encountered in cultural or historical contexts.



2.3. Procedure

An overview of the full procedure and condition structure is provided in Fig. 3 (panel A). Participants filled out a demographics section recording variables such as their age, gender and religious status. The design was presented as a “Historical Imagination study”, which instructed participants to immerse themselves in an imaginary medieval setting (S1.1. in Supplementary Materials). This framing is used as a methodological device to increase the coherence and plausibility of the task for adult participants. A difficulty in researching cultural transmission is that arbitrary laboratory actions may be approached primarily as steps required to complete a paid task, potentially obscuring the affiliative or instrumental motivations of interest. Embedding the study in a historically framed setting intends to reduce this “mere task completion” orientation and to provide a context in which the modeled actions could plausibly be construed as socially or practically meaningful.

In order to achieve immersion, participants listened to an audio recording (see <https://youtu.be/7oax0sJLqGQ>) of a guided journey into an unspecified medieval era. A sound check was administered beforehand to ensure that the audio could be heard correctly. The recording was read by a female narrator with a Southern Standard British accent (also referred to as ‘Received Pronunciation’). The first half consisted of a guided meditation (accompanied by soft background music taken from a royalty free sound database; <https://pixabay.com/music/>). Participants were then asked to mentally step into a medieval scenery, where they were given visual, olfactory and tactile descriptions paired with sound effects, such as metal work, distant conversations, a bell tower and a medieval hurdy-gurdy (volumes were adjusted to not interfere with the narrator's voice). General editing and insertion of effects was done via audio software (<https://www.audacityteam.org/>).

After the immersive exercise, participants were assigned to a team with which they played Cyberball (Williams & Jarvis, 2006). They were told that their team consisted of similar individuals based on the demographic questionnaire filled out earlier. Unbeknownst to anyone participating, the other players during Cyberball were program generated. The game consisted of controlling a virtual avatar and to pass a ball to the other players. Ostracism was manipulated across two levels: the *ostracism* condition, where players received two ball passes in the first four rounds but were then excluded from receiving any further passes, and the *control* condition where the participant received an equal number of passes evenly distributed across the duration of the game. Participants were told that ball-passing games were part of the imagined medieval setting and that the online version was included to help them situate themselves more fully in that world (S1.2. in Supplementary Materials). This framing was intended to preserve coherence between the game and the later imitation task (rather than having the game be experienced as an arbitrary interruption).

Participants then engaged in an adapted iteration of the pen-and-cups task, which has been used in the past to measure differences in copying fidelity in adult populations (Bird et al., 2007; Leighton et al., 2010). They first watched an instruction video which explained the nature of the task—putting a pen into one of two cups using one of two hand and grip choices. The instructions contained a reference to historical expert sources which were made up in order to increase the overall credibility of the cover story and to give subjects the feeling that they could potentially learn something instrumentally or socially valuable (S1.3. in Supplementary Materials).

Furthermore, both the instruction video and the stimuli used in the task were filmed with materials that fit the medieval theme. Filming was conducted in an old wood panelled room. Dark brown wooden cups were used and the pen was replaced by a light brown wooden stick. In order to make the stimuli easily discernable, a white cloth was placed underneath the cups and the model was wearing a beige tunic with laces.

A Qualtrics recording function was used, allowing direct video recording of participants' task performance.

During the setup phase, participants were asked to put water in one of the cups, depending on the condition they were in (see 2.4. *Imitation Conditions*). Different instruction videos were created for each of the conditions described below (*clear-frame* condition: <https://youtu.be/8QpZM1zHFcY>; *expert-frame* condition: <https://youtu.be/BvHi5IL5ekc>; *magic-frame* and *ritual-frame* conditions: <https://youtu.be/T0otGdplMx8>). As illustrated in Fig. 3B, the task varied along three within-subject dimensions: (a) hand choice, where the model used their left or right hand to grip the pen, (b) grip choice, where the model gripped the pen with their thumb pointed upwards or downwards, and (c) cup choice, where the model placed the pen in the left or right cup. This $2 \times 2 \times 2$ factorial combination resulted in eight unique action sequences. Each unique hand-grip-cup configuration was presented once per block, with 4 blocks being shown in total. On average, each action sequence was around 4900 ms long. The inter-trial interval was fixed at 1500 ms for all blocks. In total each block was 51 s long, with a 10 s delay between blocks. To maximize the error rate and avoid ceiling effects in task performance, participants were instructed to invert the pen before reaching out with their hand to make a cup choice.

At the end of the study, participants filled out a series of questionnaires (S2 in Supplementary Materials). They were asked to report on their overall emotional state (PANAS; Watson, Clark, & Tellegen, 1988) as well as to what degree they felt rejected during the Cyberball game using the ostracism questionnaire (Williams & Jarvis, 2006). Participants were also asked to indicate whether they were able to imagine the different aspects of their medieval journey and how much they believed the historical sources they were given, using a 10-point scale. Additionally, questions were aimed at measuring to which degree participants perceived their re-enactment to include actions that were causally opaque. Lastly, two open-ended questions were used to capture their thoughts during the Cyberball game and the copying task.

2.4. Imitation conditions

Participants were assigned to one of four conditions: the *clear-frame*, *expert-frame*, *magic-frame*, or *ritual-frame* condition, which varied the presence or absence of a physical goal, the availability of an explanation in how that goal is achieved (*resolvability*), as well as the presence or absence of causal opacity markers (for a transcript of the verbal frames see Supplementary Materials S1.3., and for a complete overview of the conditions see Fig. 3, panel C).

The *clear-frame* condition contained four instrumentality cues: (1) the mention of a filtration goal, (2) water in the starting cup, (3) a physical link between pen and cup, and (4) a given explanation. In this condition participants put water into the starting cup during the setup phase. When putting the wet pen from the starting cup into one of the two other cups, a stirring motion, which brings the pen into contact with the chosen cup, was made in order to transfer the droplets.

The *expert-frame* condition contained three instrumentality cues: (1) the mention of a filtration goal, (2) a physical link between pen and cup, and (3) an explanation which was not given but was presented as available later. It also contained one conventionality cue, as the water was located in the end-cups, rendering the causality of the sequence opaque by making it less obvious how the filtering process worked. Resolvability was cued by informing participants that the explanation for how this technique works would be given to them after the task was completed.

The *magic-frame* condition had one instrumentality cue, the mention of a filtration goal, and three conventionality cues: (1) the absence of a physical link between pen and cup, (2) the absence of water, and (3) no explanation for how the actions were supposed to work. Despite formulating a clear goal, this condition is dominated by cues that render

the causal structure opaque and more difficult to resolve, since the action was described as cleaning water from a distance and no explanation was presented as available.

The *ritual-frame* condition contained no instrumentality cues and four conventionality cues, as no water was present, the pen was not brought in contact with the cup and participants were told that this sequence was a medieval greeting (no salient physical end-goal).

2.5. Variables and data coding

Data collection resulted in 13.7 h of video footage (176 participants $\times$ 4:41 min). The full dataset was coded by two blind coders. For each trial, *cup*, *hand* and *grip* choices were coded as '1 – correct' or '0 – incorrect' depending on whether they deviated from the choice displayed by the model. These three trial-level codes were then summed into an aggregate copying-fidelity score ranging from 0 to 3, such that a participant received the maximum of 3 points on a trial if all three choices were reproduced correctly. Accordingly, a higher score was an indicator of higher copying fidelity.

Inter-rater reliability was high across all three coding dimensions (hand choice: $\kappa = 0.947$; cup choice: $\kappa = 0.928$; grip choice: $\kappa = 0.855$). Discrepancies were resolved by re-checking the original recordings against a pre-specified codebook (see section S3 in the Supplementary Materials), producing a reconciled final dataset for analysis (made available on <https://osf.io/zdy8m>). We also note that although some imitation-condition features may have been partially inferable from the movement patterns visible in the recordings, the coding was blind to the hypotheses and expected condition differences, thus reducing the likelihood of a theory-driven coding bias.

Additionally, answers to the open-ended questions were coded for key words, probing suspicions of whether the other players during Cyberball were real, as well as potential affiliative or instrumental reasoning during imitation.

3. Results

3.1. Manipulation checks

A multiple linear regression with the aggregate fidelity score as dependent variable and age, gender, ethnicity, education, and religion as predictor variables showed no effects. These variables were collected primarily for descriptive purposes and to check whether broad background characteristics were associated with copying fidelity. Further, participants who were excluded during Cyberball reported significantly higher levels of ostracism ($M = 85.41$ out of 120 possible points; $SD = 18.95$) than those in the control condition ($M = 66.47$; $SD = 18.55$), $t(174) = 6.700$, $p < .001$. Participants in the ostracism condition reported fewer ball passes ($M = 2.11$; $SD = 0.952$) than in the control condition ($M = 6.26$; $SD = 2.926$), $t(174) = -12.644$, $p < .001$, as well as significantly more negative emotions ($M = 24.73$; $SD = 7.68$) than in the control condition ($M = 17.39$; $SD = 6.82$), $t(174) = 6.700$, $p < .001$.

Participants also reported immersion scores significantly above the scale midpoint of 25 ($M = 34.32$ out of 50 possible points; $SD = 8.82$), $t(175) = 14.021$, $p < .001$, suggesting that the historical imagination procedure engaged participants to a meaningful degree. A factorial ANOVA reveals no differences in immersion between social conditions, $F(1, 168) = 0.776$, $p = .380$, and none between imitation conditions, $F(3, 168) = 0.910$, $p = .438$, indicating that immersion was constant across conditions. Participants were also asked how much they believed the historical sources they were given. Credibility ratings were also significantly above the scale midpoint of 3 ($M = 3.95$ out of 5; $SD = 0.85$), $t(174) = 14.715$, $p < .001$, indicating that participants found the historical sources believable. A factorial ANOVA reveals no differences between social conditions, $F(1, 168) = 2.246$, $p = .136$, and none between imitation conditions, $F(3, 168) = 0.574$, $p = .633$, indicating that the level of credibility was constant across conditions.

Lastly, participants were also prompted to share the thoughts they had during the Cyberball task. 41 participants (23.3%) expressed doubts that the Cyberball task was a real game (25 in the ostracism and 16 in the control condition). A 2×2 Factorial ANOVA with social conditions (ostracism vs. control) and suspicion (suspicious vs. unsuspicious) as independent variables and average scores on the reported ostracism questionnaire as dependent variable, revealed no effect of suspicion on reported feelings of rejection, $F(1, 172) = 1.269$, $p = .263$.

3.2. The effects of ostracism and learning cues on imitation

3.2.1. Model diagnostics and robustness

A 2×4 factorial ANOVA with social conditions (ostracism vs. control) and imitation conditions (*clear-frame*, *expert-frame*, *magic-frame*, *ritual-frame*) as between-subjects variables, and the aggregated copying fidelity score as dependent variable was used. Residual diagnostics indicated heteroscedasticity across several factor-level combinations and modest departures from normality. Three observations showed elevated Cook's distances, but none exceeded conventional influence thresholds. Importantly, the pattern of results was unchanged when analyses were repeated with these observations removed. Given the presence of unequal variances, we estimated the model using heteroskedasticity-consistent (HC3) standard errors and we report these results together with robust pairwise contrasts below. These HC3 results are consistent with the classical ANOVA, indicating that the observed interaction between social and imitation conditions is not contingent on parametric assumptions. To further evaluate the stability of these effects, we conducted a nonparametric bootstrap (5000 resamples). Inspection of the bootstrap percentile confidence intervals provided convergent evidence that effects are robust to deviations from normality and the conclusions we draw from the data therefore remain unchanged.

H1. Ordered differences across imitation conditions.

To test the preregistered H1 prediction more directly, we computed a planned contrast corresponding to the expected ordering *clear-frame* < *expert-frame* < *magic-frame* < *ritual-frame*. Averaged across social conditions, this ordered contrast was significant, estimate = 1.020, $SE = 0.434$, 95% CI [0.159, 1.870], $t(168) = 2.340$, $p = .021$. When the same ordered contrast was examined separately by social condition, it was not significant in the control group, estimate = -0.349 , $SE = 0.649$, 95% CI [-1.630 , 0.931], $t(168) = -0.539$, $p = .591$, and significant in the ostracized group, estimate = 2.381, $SE = 0.577$, 95% CI [1.240, 3.520], $t(168) = 4.125$, $p < .001$. This indicates that the preregistered ordering was expressed primarily under ostracism.

H2. Ostracism selectively amplifies copying in conventionally framed conditions.

The HC3 ANOVA reveals a significant interaction between the social conditions (ostracism vs. control) and the imitation conditions (*clear-frame*, *expert-frame*, *magic-frame*, *ritual-frame*), $F(3, 168) = 3.753$, $p = .012$, $\eta_p^2 = 0.066$ (Fig. 4.A.). Robust pairwise contrasts, with Bonferroni-adjusted p -values, revealed that in the *magic-frame* condition, ostracized participants copied actions significantly more accurately ($M = 2.38$, $SE = 0.08$) than those that were not ostracized during Cyberball (control; $M = 1.79$, $SE = 0.14$), $p = .0005$, mean difference = 0.590, 95% CI = [0.264, 0.915]. Equally, when copying actions in the *ritual-frame* condition, ostracized participants copied more accurately ($M = 2.54$, $SE = 0.07$) than the control group ($M = 1.87$, $SE = 0.16$), $p = .0003$, mean difference = 0.665, 95% CI = [0.307, 1.023]. In contrast, ostracism showed no effect on copying fidelity in the *clear-frame* and *expert-frame* conditions.

To test the preregistered H2a–H2d logic more directly, we additionally computed planned interaction contrasts comparing the size of the ostracism effect in each condition against the *clear-frame* baseline, where no ostracism effect was predicted. These contrasts showed that

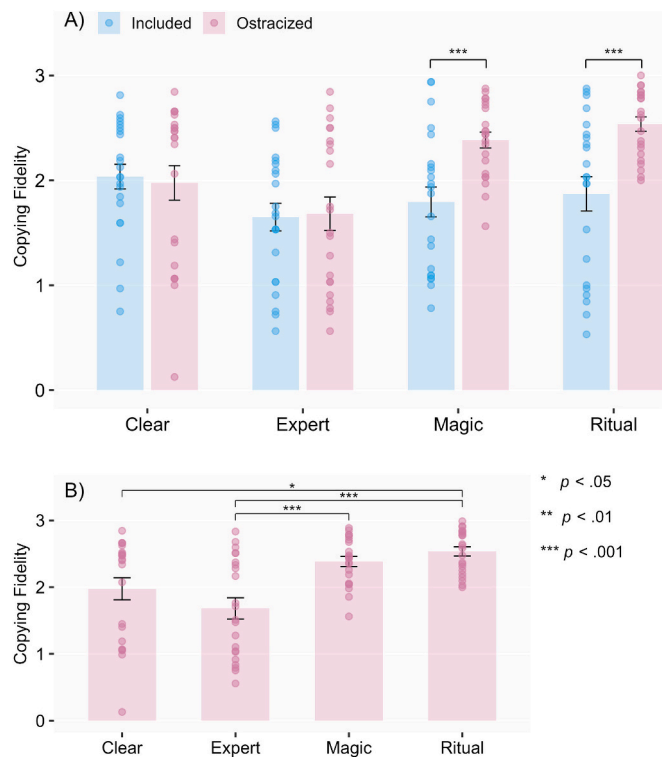


Fig. 4. Mean copying fidelity for each imitation condition (clear-frame, expert-frame, magic-frame, and ritual-frame) separately for ostracized (purple) and non-ostracized control (blue) participants. Bars represent condition means, error bars indicate standard error, and dots show individual participant observations. The upper panel (A) displays the full interaction between imitation conditions and social conditions (ostracism vs. control). Differences between ostracized and control participants emerge most clearly in the magic-frame and ritual-frame conditions. The lower panel (B) shows copying fidelity within the ostracized condition, illustrating effects across imitation conditions. The y-axis reflects the bounded copying fidelity score (0–3).

the ostracism effect was significantly larger in the *magic-frame* condition than in the *clear-frame* condition, mean difference in ostracism effects = 0.649, 95% CI [0.009, 1.290], $p = .031$, and significantly larger in the *ritual-frame* condition than in the *clear-frame* condition, mean difference in ostracism effects = 0.724, 95% CI [0.059, 1.390], $p = .028$. By contrast, the ostracism effect in the *expert-frame* condition did not differ from that in the *clear-frame* condition, mean difference in ostracism effects = 0.092, 95% CI [−0.623, 0.808], $p = .756$. Ordering contrasts among the non-clear conditions (*ritual-* vs. *magic-* vs. *expert-frame*) did not reach significance after correction (all $ps \geq 0.074$).

3.2.2. Follow-up comparisons within each social condition

Further comparisons were run to determine how the imitation conditions differed within each social condition (ostracism vs. control). In the control condition, copying fidelity did not differ between imitation conditions. In the ostracized condition, participants in the ritual-frame condition ($M = 2.54$, $SE = 0.07$) were more accurate than those in the clear-frame condition ($M = 1.98$, $SE = 0.17$), $p = .018$, mean difference = 0.560, 95% CI [0.072, 1.047] (Fig. 4B). Those in the ritual-frame condition were also more accurate than those in the expert-frame condition ($M = 1.68$, $SE = 0.16$), $p = .0001$, mean difference = 0.854, 95% CI [0.380, 1.327]. Participants in the magic-frame condition ($M = 2.38$, $SE = 0.08$) likewise copied more accurately than those in the expert-frame condition ($M = 1.68$, $SE = 0.16$), $p = .0009$, mean difference = 0.702, 95% CI [0.219, 1.184]. The remaining contrasts within the ostracized condition (*clear-frame* vs. *expert-frame*, *clear-frame* vs. *magic-frame*, and *magic-frame* vs. *ritual-frame*) were non-significant.

H3 and H4. Reported feelings of ostracism, negative affect and immersion.

Across the entire sample, reported feelings of ostracism were positively associated with copying fidelity, $r(176) = 0.160$, $p = .034$. To test the preregistered H3 prediction, we examined whether reported feelings of ostracism predicted copying fidelity differently across imitation conditions. A robust regression of copying fidelity on reported ostracism, imitation condition, and their interaction showed no interaction between reported ostracism and imitation conditions, $F(3, 168) = 0.561$, $p = .641$. Condition-specific slopes were likewise non-significant. Reported negative affect did not predict copying fidelity overall, $F(1, 168) = 0.655$, $p = .419$, and did not interact with imitation conditions, $F(3, 168) = 1.282$, $p = .281$. Condition-specific slopes were likewise non-significant.

To test the preregistered H4 prediction, we fitted a model including immersion, social condition, imitation condition, and their interactions. This analysis did not reveal any interactions between immersion and the social or imitation conditions, $F(3, 160) = 1.816$, $p = .241$.

3.2.3. Exploratory analysis of performance heterogeneity

Because Fig. 4 suggested not only mean differences but also differences in variability across cells, we conducted an exploratory analysis across the full 2×4 design. For this purpose, we computed each participant's absolute deviation from the median copying-fidelity score of their own cell and used this as an index of performance heterogeneity. A robust factorial analysis of these deviations revealed a significant interaction between social condition and imitation condition, $F(3, 168) = 5.915$, $p = .001$, indicating that the effect of ostracism differed across action types.

Follow-up contrasts showed that performance heterogeneity was significantly lower under ostracism (compared to control) in the *magic-frame* condition, mean difference = -0.274 , 95% CI [−0.456, −0.093], $p = .003$, and in the *ritual-frame* condition, mean difference = -0.344 , 95% CI [−0.552, −0.136], $p = .001$, but not in the *clear-frame* or *expert-frame* conditions. Descriptively, mean absolute deviation was lowest in the ostracized *magic-frame* ($M = 0.284$) and ostracized *ritual-frame* ($M = 0.280$) cells, compared with their respective control baselines (*magic-frame*: $M = 0.558$; *ritual-frame*: $M = 0.624$). We interpret this cautiously, as performance heterogeneity was not a preregistered outcome, but note that it is consistent with the idea that under ostracism threat, conventionally weighted action sequences may be copied not only more faithfully, but also to similar degrees of faithfulness across participants, indicating that performance homogeneity may be important for ritual transmission.

3.2.4. The level of deliberateness

In order to gauge how conscious stance adoption is, we asked participants to answer questions about the causal opacity of the actions sequences they imitated. Analysis of these data was also exploratory as no pre-registered predictions were made. A One-Way ANOVA with imitation conditions (*clear-frame*, *expert-frame*, *magic-frame*, *ritual-frame*) as between-subjects variables, and scores on the causal opacity questions as dependent variable revealed no effect of the imitation conditions, $F(3, 172) = 0.201$, $p = .896$, suggesting that participants did not consciously see any differences in terms of how causally opaque the actions seen during imitation were. We also asked participants to share the thoughts they had during the imitation task in an open-ended question. Our analysis indicates that answers were predominantly idiosyncratic in nature, with no mentions of affiliative concerns, while overall mentions of achieving a goal were rare (10.2%) and did not differ across clear, expert, magic and ritual conditions, $X^2(1, N = 176) = 2.576$, $p = .109$.

4. Discussion

4.1. Main findings

The present study tested whether ostracism threat changes copying fidelity as a function of how an observed action sequence is framed. Our main finding is an interaction between ostracism and the degree to which actions were framed to be causally opaque, goal-demoted, and resolvable. Ostracized participants copied more faithfully than non-ostracized participants in the *magic-frame* and *ritual-frame* conditions, where conventionality cues outweighed instrumentality cues, whereas no ostracism effect emerged in the *clear-frame* or *expert-frame* conditions, where instrumentality was made more salient. Planned interaction contrasts confirmed this pattern: the ostracism effect was significantly larger in both the *magic-frame* and *ritual-frame* conditions than in the *clear-frame* baseline, while the *expert-frame* condition did not differ from *clear-frame*. This supports the core prediction that ostracism threat selectively amplifies copying when action sequences are weighted toward conventionality and irresolvability. At the same time, there was no evidence for a fully graded hierarchy in the size of the ostracism effect across all non-clear conditions (*expert-*, *magic-*, and *ritual-frame*), indicating that ostracism threat generally increases copying when actions are framed as conventional.

The preregistered H1 contrast addresses a related but distinct question: whether copying fidelity itself follows the predicted ordering *clear-frame* < *expert-frame* < *magic-frame* < *ritual-frame*. This ordered contrast received support across the whole sample with the effect being strongly present in the ostracized group and absent in the control group. Thus, the predicted differentiation emerged most clearly when social belonging was threatened, compared to the non-ostracized control condition. We also conducted an exploratory analysis of performance heterogeneity which adds a further layer to our results: Under ostracism, participants in the *magic-frame* and *ritual-frame* conditions were not only more faithful in their copying, but also more similar to one another in their level of copying fidelity (compared to the *clear-frame* or *expert-frame* conditions). We therefore cautiously note the possibility that, under ostracism threat, conventionally weighted action sequences may invite not only higher fidelity, but also greater homogeneity of performance. Such homogeneity may be important for ritual transmission, where similar levels of energetic investment can lead to fewer deviations from the script.

Furthermore, the difference between the *expert-frame* and *magic-frame* conditions is especially informative, as it allows to compare predictions generated by BST (Jagiello et al., 2022b) to other more goal-focused accounts (Dubourg et al., 2022; Hong, 2022). Both involved causally opaque action sequences with an end-goal, but only the *magic-frame* condition produced a reliable increase in copying fidelity under ostracism. This may suggest that despite the presence of an end-goal, an action sequence that predominantly signals conventionality and *irresolvability* is transmitted faithfully when affiliative motivations are made salient. While the presence of an explicit end-goal likely constitutes an important learning cue, our findings show that it is not sufficient on its own to determine the learning profile with which actions are copied during transmission. Rather, it appears that learners are sensitive to a large set of cues that are tied to both socially and physically relevant outcomes.

The secondary set of preregistered analyses was less supportive. Reported feelings of ostracism were positively associated with copying fidelity across the full sample, but not for specific imitation conditions. Negative affect did not predict copying fidelity, neither across the whole sample, nor in specific imitation conditions. Even though we remain cautious about the role of generalized affect, this difference might indicate that our main effect is not easily reduced to non-specific negative emotion (as discussed below). Moreover, the prediction that higher immersion would be associated with copying fidelity among

ostracized participants in the opaque conditions did not find support. The main evidential weight of our findings therefore rests on the planned H1 and H2 findings rather than on individual-difference mediators (H3 and H4).

We also examined the degree of deliberateness with which social learners adopt either stance by asking participants to report on the causal opacity of the action sequences and to describe the thoughts they experienced during the imitation task. We did not pre-register a prediction in favour of either conscious or automatic stance-adoption as BST generally remains agnostic toward the precise cognitive architecture of learning motivations (Jagiello et al., 2022a). Here, we observed no consistent patterns of goal-focused or affiliation-focused reasoning, suggesting limited conscious access to stance-related motivations. Given that our self-report measures may have lacked the sensitivity needed to capture participants' reasoning, future work will need more direct tests of whether stance adoption is deliberative, automatic or mixed.

Taken together, the findings support a *learner-centric* account of cultural transmission, where goal-directed but causally opaque sequences may be copied differently depending on whether learners interpret them as instrumentally resolvable procedures or as conventionally meaningful practices. Under ostracism threat, actions framed in the latter way are copied more faithfully and more homogeneously. We now discuss the limitations, noting that some aspects of our design should invite guarded enthusiasm in how the current results are interpreted.

5. Limitations and future directions

A number of limitations apply to our results. First, although the target sample size was preregistered on the basis of an a priori power analysis informed by previous work and piloting, the study remains modestly powered for fine-grained interaction contrasts. The main ostracism pattern (H2) was robust across non-parametric testing, influence checks and bootstrapping, but estimates for individual-difference analyses (H3 and H4) should still be treated cautiously – for instance, it remains a possibility that despite the null findings, reported feelings of ostracism or negative affect still play a role during imitation.

Second, the Cyberball task allowed us to use ostracism to make affiliative motivation salient, while the non-ostracized condition in which participants received an equal number of passes served as a baseline control. We regard the induction of affiliative motivations via ostracism as a methodological necessity given that a short laboratory procedure cannot easily reproduce genuine group membership or natural belonging based on shared history. This means that the study speaks most directly to the effect of threatened belonging, not to all forms of ordinary in-group learning. Rather than simulating naturalistic group entry, the present design focuses on how threatened belonging heightens affiliative motivation, which may then increase sensitivity to, or the salience of, affiliative and conventional cues in subsequent learning instances. Importantly, this does not require participants to consciously infer that copying the demonstrated action would allow them to rejoin the Cyberball players. Social exclusion can increase affiliative mimicry beyond the original exclusion episode (Lakin, Chartrand, & Arkin, 2008). Consistent with this, our own exploratory measures did not suggest that participants were consciously theorizing about stance adoption during the task.

Moreover, the absence of a ritual-frame effect in the control condition further clarifies the scope of the present findings: ritual-frame cues did not increase copying fidelity in the control condition, suggesting that goal salience and resolvability do not determine copying fidelity on their own. Instead, they appear to be one set of cues whose influence depends on the learner's broader motivational and social context, consistent with BST as a cue-weighting account. Future research should therefore also test whether similar effects can be produced by positive group induction rather than exclusion. If established group belonging produces similar

increases in fidelity for conventionally framed actions, this would strengthen the claim that the ritual stance is the key driver of the observed effects.

Third, the historical framing and the recruitment criteria may limit generalizability beyond this particular sample. Again, we regard these design choices as necessary methodological trade-offs in studying cultural transmission in adults: When laboratory tasks are made highly abstract, decontextualized, or overtly game-like in the name of experimental control, they may become equally limited in generalizability, insofar as participants approach them primarily as arbitrary steps toward payment rather than as socially or instrumentally meaningful behaviour. Our design therefore sought a middle ground between ecological plausibility and experimental control. While the historically framed context may introduce some constraints, we do not think a more sterile laboratory format would be able to provide a robust window onto the motivations involved in social learning. Nonetheless, the effects we report here need to be replicated across other samples and learning contexts.

Fourth, resolvability was operationalized through explicit framing cues rather than allowed to emerge spontaneously through participants' own discovery. Our principal aim was to test whether cues suggesting that an action is causally irresolvable can influence how it is copied, rather than simulating the entire cognitive process by which learners arrive at conclusions about resolvability. Even so, it means that perceived resolvability cannot be cleanly separated here from other framing-related cues, such as authority. Accordingly, the present findings are best understood as evidence that cue-guided interpretations of resolvability matter for copying, but future studies should manipulate these components more separately.

Fifth, the design does not fully disentangle affiliative motivation from other consequences of ostracism (Albath, Bogatyreva, Büttner, & Greifeneder, 2025). Precise copying in the *magic-frame* and *ritual-frame* conditions may have been socially meaningful, but it may also have provided structure, predictability or anxiety reduction following exclusion (Lang, Krátký, & Xygalatas, 2022). Negative affect did not explain copying fidelity in the present data, which argues against a wholly non-specific affect account, but stronger control conditions are needed in future work, for instance by comparing ostracism to randomized task failure or loss-of-control manipulations (Landau, Kay, & Whitson, 2015; Whitson & Galinsky, 2008) in order to determine whether the effect is specific to social exclusion or reflects a broader response to distress and uncertainty.

Lastly, a further question for future research concerns the developmental and cultural origins of these learning profiles. The present study was not designed to test whether stance-related differences reflect domain-specific, biologically prepared capacities or culturally acquired learning strategies. Nevertheless, this remains an important issue, particularly in cross-cultural and developmental designs that vary pedagogical input and socialization history (Clegg et al., 2021; Clegg & Legare, 2016a; Clegg, Wen, & Legare, 2017; Clegg, Wen, & Rawlings, 2022; Rybanska, McKay, Jong, & Whitehouse, 2018).

6. Conclusion

Ritual and instrumental stances both play a fundamental role in cultural learning. However, we argue that the distinction is in the eye of the beholder rather than an objective description of distinct types of behaviour. Cultural practices (including magic and its diffusion) may be interpreted very differently depending on who performs them, who observes their performance, and the context in which those observations take place. Ultimately, our framework seeks to provide insights into the properties, conditions and general affordances of the social learning interactions that make one type of interpretation more likely than the other. In doing so, we hope to contribute to ongoing efforts to understand how cultural practices come to be interpreted, learned and sustained as meaningful parts of human social life.

CRedit authorship contribution statement

Robert Jagiello: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Harvey Whitehouse:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

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Declaration of competing interest

Nothing to declare.

Data availability

Data are available on <https://osf.io/zdy8m>.

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